



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI ISWARYA TRADERS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

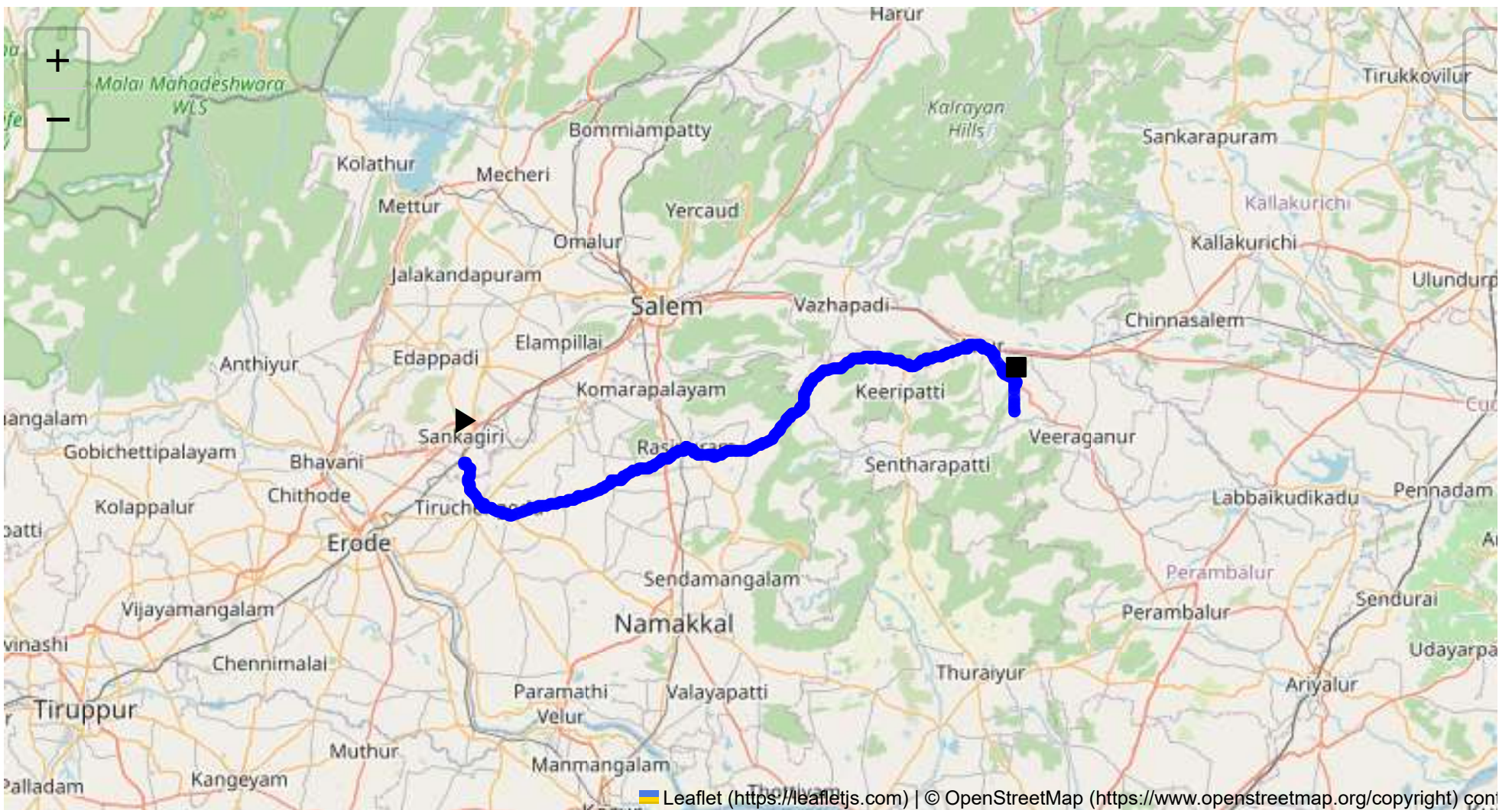
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 109.90 km
Estimated Duration: 2.6 hours
Adjusted Duration (Heavy Vehicle): 3.2 hours
Start: (11.4381, 77.8734)
End: (11.510717, 78.643089)



Welcome to the Journey Risk Management Study

Route Overview

The route starts at Sangagiri, Tamil Nadu (CVQF+23W), and ends at Higoli, Tamil Nadu (GJ6V+76). It covers approximately 109.90 kilometers, passing through several towns, including Tiruchengode, Kavundampalayam, Rasipuram, and Salem. This route features a mix of state highways and local roads.

Weather Conditions

Tamil Nadu generally experiences a tropical climate, with high temperatures throughout the year. The hot season could lead to tire blowouts and other heat-related issues. The monsoon season (June to September) brings heavy rains, causing potential flooding and slippery roads, which could impact visibility and driving conditions.

Traffic Patterns

Traffic intensity varies, with peak congestion in urban areas like Salem and during morning (8 AM - 10 AM) and evening (5 PM - 8 PM) rush hours. Market days can also increase local traffic, particularly in smaller towns along the route.

Road Quality and Infrastructure

The main highways like SH 79 are generally in good condition, but smaller roads in rural areas may have uneven surfaces and limited signage. Be wary of potholes and narrow passages, especially in older sections of towns.

Alternative Routes

In case of emergencies, consider routes like:

- SH 153 via Erode for a more direct connection toward northern destinations
- Local roads through Namakkal for avoiding major towns These alternatives may have limitations in terms of road quality and increase travel time.

Local Regulations

Transporting hazardous materials requires adherence to strict protocols, including allowable transport times, necessary permits, and observance of safety standards. It's also crucial to follow speed limitations (around 40-60 km/h in towns and 60-80 km/h in rural stretches).

Historical Incidents

There are few documented major incidents involving heavy vehicles along this route; however, these can be attributed to human error and adverse weather, underscoring the need for cautious driving practices.

Environmental Considerations

Sensitive areas include agricultural fields and water bodies near the route, requiring pollution-prevention practices. Minimizing noise pollution near schools and residential areas is also crucial.

Communication Coverage

The major highways and towns generally have strong cellular coverage. However, some rural stretches may experience dropped signals or weak reception. Ensure offline maps and communication devices are

prepared.

Emergency Response Times

Urban areas like Salem have quicker response times (approx 15-30 minutes), while rural segments may experience delays (up to an hour) due to distance from emergency services.

Overall Risk Assessment

This route poses moderate risk to heavy vehicles carrying hazardous materials. Key hazards include erratic weather, potential for congestion, and road quality variations. Preparedness for these conditions, with an emphasis on careful planning, can mitigate many risks.

Overall, the emphasis must be on vigilant driving, route familiarity, and adhering strictly to safety and regulatory protocols to ensure safe transit of hazardous materials through this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44019, 77.87547	15 KM/Hr
3	Turn	Medium	11.39685, 77.88861	30 KM/Hr
4	Turn	Medium	11.38382, 77.89897	30 KM/Hr
5	Turn	High	11.38188, 77.90257	15 KM/Hr
6	Blind Spot	Blind Spot	11.37830, 77.90160	10 KM/Hr
7	Turn	Medium	11.41361, 78.08073	30 KM/Hr
8	Blind Spot	Blind Spot	11.41453, 78.08062	10 KM/Hr
9	Turn	High	11.43240, 78.12064	15 KM/Hr
10	Turn	High	11.45586, 78.18045	15 KM/Hr
11	Blind Spot	Blind Spot	11.46035, 78.18534	10 KM/Hr
12	Turn	High	11.45904, 78.18556	15 KM/Hr
13	Turn	Medium	11.45132, 78.19979	30 KM/Hr
14	Turn	Medium	11.44994, 78.22138	30 KM/Hr
15	Turn	Medium	11.44919, 78.22291	30 KM/Hr
16	Turn	High	11.44806, 78.22308	15 KM/Hr
17	Turn	Medium	11.44761, 78.22386	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
18	Turn	Medium	11.47210, 78.30450	30 KM/Hr
19	Turn	Medium	11.49813, 78.32398	30 KM/Hr
20	Turn	Medium	11.50501, 78.33083	30 KM/Hr
21	Turn	Medium	11.50776, 78.33301	30 KM/Hr
22	Turn	Medium	11.50861, 78.33957	30 KM/Hr
23	Turn	Medium	11.51153, 78.34384	30 KM/Hr
24	Turn	Medium	11.51854, 78.34807	30 KM/Hr
25	Turn	Medium	11.54816, 78.35606	30 KM/Hr
26	Turn	Medium	11.56708, 78.37800	30 KM/Hr
27	Turn	Medium	11.58164, 78.41774	30 KM/Hr
28	Turn	Medium	11.58329, 78.44366	30 KM/Hr
29	Turn	Medium	11.57840, 78.48417	30 KM/Hr
30	Turn	Medium	11.57093, 78.49976	30 KM/Hr
31	Turn	Medium	11.57100, 78.50071	30 KM/Hr
32	Turn	Medium	11.57840, 78.51270	30 KM/Hr
33	Turn	Medium	11.59100, 78.55627	30 KM/Hr
34	Turn	Medium	11.59237, 78.55745	30 KM/Hr
35	Turn	Medium	11.59461, 78.56846	30 KM/Hr
36	Turn	High	11.59744, 78.57347	15 KM/Hr
37	Turn	Medium	11.59589, 78.57440	30 KM/Hr
38	Turn	High	11.59538, 78.57598	15 KM/Hr
39	Turn	High	11.59892, 78.57708	15 KM/Hr
40	Turn	Medium	11.59982, 78.58078	30 KM/Hr
41	Turn	High	11.60136, 78.58233	15 KM/Hr
42	Turn	High	11.59957, 78.59646	15 KM/Hr
43	Turn	High	11.59852, 78.59670	15 KM/Hr
44	Turn	Medium	11.59631, 78.60331	30 KM/Hr
45	Turn	Medium	11.59376, 78.60597	30 KM/Hr
46	Turn	High	11.59354, 78.60992	15 KM/Hr
47	Turn	Medium	11.59282, 78.61016	30 KM/Hr
48	Turn	Medium	11.56766, 78.62561	30 KM/Hr
49	Turn	Medium	11.56478, 78.62622	30 KM/Hr
50	Turn	High	11.55475, 78.64297	15 KM/Hr
51	Turn	Medium	11.55161, 78.64379	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
52	Turn	Medium	11.55101, 78.64456	30 KM/Hr
53	Turn	Medium	11.55016, 78.64494	30 KM/Hr
54	Turn	High	11.51094, 78.64274	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
12	hospital	Sudha Hospital	11.4555773, 78.1760125	30 km/h	Medium
13	hospital	Government Hospital Blood Bank, Rasipuram	11.4612082, 78.1849249	30 km/h	Medium
14	hospital	M.R. Hospital, Namakkal	11.4613462, 78.1861702	30 km/h	Medium
15	hospital	dr.thangavel dch hospital	11.4619955, 78.1849181	30 km/h	Medium
16	police	Police Station Rasipuram	11.4617825, 78.1847411	30 km/h	Medium
17	hospital	Government Hospital, Rasipuram	11.4612987, 78.1851369	30 km/h	Medium
18	hospital	Rasi Hospital	11.4618368, 78.1855695	30 km/h	Medium
19	hospital	surya hospital	11.4622163, 78.1840544	30 km/h	Medium
20	hospital	Manickkam Hospital	11.4568944, 78.1926061	30 km/h	Medium
21	hospital	R K Eye Care Center	11.4589079, 78.1913596	30 km/h	Medium
23	hospital	Snathi Hospital	11.4564245, 78.1932485	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
24	hospital	Sugam Hospital,Rasipuram	11.4553007, 78.1936221	30 km/h	Medium
25	hospital	Government Hospital, Namagiripettai	11.4569484, 78.265263	30 km/h	Medium
26	clinic	Dr. Saravanakumar Clinic	11.5842002, 78.4441223	30 km/h	Medium
27	hospital	Government Hospital	11.572472, 78.497407	30 km/h	Medium
29	hospital	Abi Hospital	11.59319, 78.560129	30 km/h	Medium
32	hospital	Sundaram Multi Speciality Hospital	11.600633, 78.5948143	30 km/h	Medium
33	clinic	RK Clinic	11.59969, 78.5941386	30 km/h	Medium
34	clinic	Attur ENT and Dental Care Centre	11.5991073, 78.5962503	30 km/h	Medium
35	hospital	Kalyani Memorial Hospital	11.599983, 78.596223	30 km/h	Medium
36	clinic	Mother's Ortho Clinic	11.6002438, 78.594053	30 km/h	Medium
37	hospital	Arun Hospital	11.599848, 78.596343	30 km/h	Medium
38	hospital	Chiranjeevi Hospital	11.5998283, 78.594309	30 km/h	Medium
39	clinic	Sri Vinayaga Physiotherapy Clinic	11.598134, 78.597401	30 km/h	Medium
40	hospital	Sri Andavhar Nursing Home	11.6010351, 78.5974614	30 km/h	Medium
41	clinic	SS Clinic	11.5975084, 78.5979036	30 km/h	Medium
42	clinic	Bawaa Clinic	11.597934, 78.597143	30 km/h	Medium
43	clinic	Ramakrishna Nursing Home	11.5988349, 78.5988615	30 km/h	Medium
44	hospital	Gokulam Hospital	11.597356, 78.598403	30 km/h	Medium
45	hospital	Kumar Eye Hospital	11.5978065, 78.5984006	30 km/h	Medium
46	hospital	Rk Memorial Hospital	11.5975972, 78.5993824	30 km/h	Medium
47	hospital	Anugraha Hospital	11.5963566, 78.5941161	30 km/h	Medium
48	hospital	Sakthi Sivam Hospital	11.5959889, 78.5941723	30 km/h	Medium
49	hospital	Government Hospital Attur	11.59709, 78.601343	30 km/h	Medium
50	hospital	N.S. Kumudham Hospital	11.59626, 78.599512	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
51	hospital	Abhi Eye Hospital	11.5980207, 78.6003698	30 km/h	Medium
52	hospital	Government Hospital	11.597079, 78.600619	30 km/h	Medium
53	hospital	Panneerselvam Hospital	11.598067, 78.599939	30 km/h	Medium
54	clinic	Dr. Ramdoss Clinic	11.595518, 78.600865	30 km/h	Medium
55	hospital	Sri Dhanvantari Children Hospital	11.5991194, 78.5999856	30 km/h	Medium
56	hospital	Arul Gandhi Physiotherapy Hospital	11.5979315, 78.6003262	30 km/h	Medium
57	hospital	St. Joseph Hospital	11.5990918, 78.6003644	30 km/h	Medium
58	clinic	G.V. Homeo Clinic	11.596099, 78.600999	30 km/h	Medium
59	hospital	Lakshitha Hospital	11.5952715, 78.6094785	30 km/h	Medium
60	hospital	Mugamadhu Absal Hospital	11.51225, 78.643016	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium
7	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium
8	school	Vidya madir school	11.4356335, 78.1381408	30 km/h	Medium
9	college	Thiruvalluvar Govt arts college	11.4449352, 78.1595562	30 km/h	Medium
10	school	Venkateswara School	11.4468258, 78.1595761	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
11	school	Venkateswara Middle school	11.4472063, 78.1596191	30 km/h	Medium
22	school	Govt. Hr. Sec. School	11.4585324, 78.1933666	30 km/h	Medium
28	school	Rasi Matric HR Sec School	11.5760027, 78.5062808	30 km/h	Medium
30	school	Government Boys Higher Secondary School	11.598743, 78.5885472	30 km/h	Medium
31	school	Government Boys Higher Secondary School	11.599149, 78.5888931	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44019, 77.87547



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39685, 77.88861



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38382, 77.89897



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38188, 77.90257



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.37830, 77.90160



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41361, 78.08073



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.41453, 78.08062



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.43240, 78.12064



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45586, 78.18045



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.46035, 78.18534



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45904, 78.18556



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45132, 78.19979



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44994, 78.22138



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44919, 78.22291



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44806, 78.22308



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44761, 78.22386



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.47210, 78.30450



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49813, 78.32398



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.50501, 78.33083



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.50776, 78.33301



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.50861, 78.33957



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.51153, 78.34384



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.51854, 78.34807



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.54816, 78.35606



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56708, 78.37800



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58164, 78.41774



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58329, 78.44366



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57840, 78.48417



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57093, 78.49976



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57100, 78.50071



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57840, 78.51270



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59461, 78.56846



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.59744, 78.57347



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59589, 78.57440



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.59538, 78.57598



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.59892, 78.57708



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59982, 78.58078



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.60136, 78.58233



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.59957, 78.59646



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.59852, 78.59670



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59631, 78.60331



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59376, 78.60597



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.59354, 78.60992



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59282, 78.61016



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56766, 78.62561



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56478, 78.62622



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.55475, 78.64297



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55161, 78.64379



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.55101, 78.64456



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.55016, 78.64494



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.51094, 78.64274
